Project Title:

Conversion of post-combustion gas from cement manufacturing to syngas by electrochemical CO_2 capture and reduction

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February monthly report

1. **Tasks list for February**

- Long-term stability under high CO_2 conversion conditions (diluted and pure CO_2 conditions with lower flow rates)
- Long-term stability under SFG conditions at 200mA/cm^2

2. **Experimental Procedure**

Long-term stability of CO_2 electrolysis

A 5 cm^2 MEA electrolyzer was used to evaluate the CO2RR performance under modulated operating conditions. The Ni foam, bipolar membrane (BPM), and different cathode catalysts were employed as the anode, membrane, and cathode electrode, respectively. 500 ml of 0.5 KOH was recirculated and used throughout the experiment as the anolyte at 8 ml/min. The KOH electrolyte was refreshed every 24h, while the NiSAC cathode was also flushed with 10ml of DI water. The humidified CO_2 gas stream was introduced into the cathodic inlet under total flow rates of 10 ml/min. Meanwhile, the cathodic outlet was directly injected into an online gas-chromatography for gas products analysis.

3. **Results**

Long-term CO_2 electrolysis of Ni0.5SAC under 20ml SFG with 0.25M KOH and IrO_2-TiPt mesh at 100-200mA/cm^2

This figure presents data from CO_2 electrolysis experiments conducted at current densities of 100 mA/cm² (left) and 200 mA/cm² (right) using a 20 ml SFG feed (15% CO_2, 5% O_2, 80% N_2) in 0.25M KOH with an IrO_2-TiPt anode. The graphs show the Faradaic efficiency (FE) for CO (green) and H_2 (pink) production, along with cell voltage over time. At 100 mA/cm², CO formation remains low (< 20%) while H_2 production gradually increases, and the single-pass CO_2 utilization (SPU) is 28%, indicating poor efficiency due to higher ORR activity. In contrast, at 200 mA/cm², CO formation is initially higher but decreases slightly, while H_2 production remains relatively stable, leading to a much higher SPU of 92% with total FE(CO+H2) around 80%, suggesting significantly improved efficiency in terms of suppressing ORR activity. The cell voltage rises in both cases, from ~3V to 4V over 24 hours at 100 mA/cm² and from ~3.5V to 4.5V within 2 hours at 200 mA/cm². These results indicate that increasing the current density greatly improves power utilization efficiency, although the long-term stability of CO_2 electrolysis must be further improved.

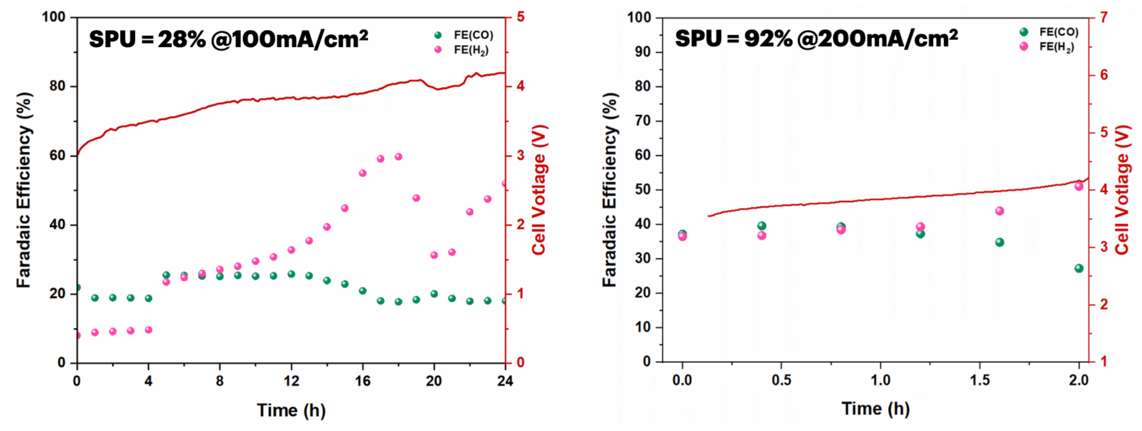

Figure 1. CO_2 electrolysis under 20ml of SFG (15%CO_2,5%O_2,80%N_2) feed with 0.25M KOH and IrO_2-TiPt anode at 100 and 200mA/cm^2

This figure illustrates the effect of different diluted 15%CO_2 flow rates (100 ml/min, 50 ml/min, and 30 ml/min) at a constant current density of 200 mA/cm² on CO_2 electrolysis performance. As the total flow rate of 15%CO_2/85%N_2 decreases from 100 ml/min to 30 ml/min, the FE for CO marginally decreases, indicating poor CO selectivity under lower CO2 concentration, while the FE for H_2 remains relatively stable with a slight decline. The single-pass utilization (SPU) improves significantly, increasing from 36% at 100 ml/min to 95% at 30 ml/min with a short operational duration. This trend suggests a tradeoff between SPU and stability of the overall cell system under various CO_2 supply.

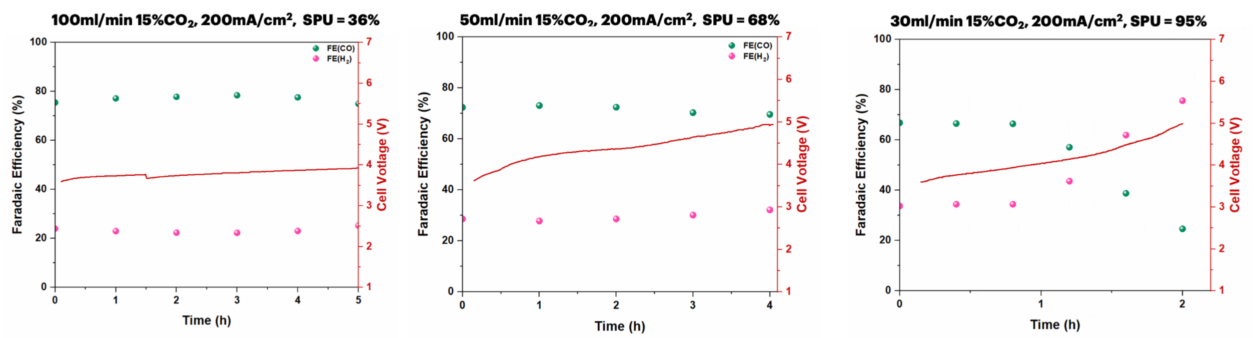

Figure 2. CO_2 electrolysis under different flow rates of 15%CO_2/85%N_2 feed with 0.25M KOH and IrO_2-TiPt anode at 200mA/cm^2

Influence of CO_2 supply on single-pass utilization (SPU) at 200mA/cm^2

In order to determine the carbon efficiency of the CO_2 electrolysis system, we study the impact of CO_2 supply on selectivity and SPU. At 150 ml/min, CO_2 utilization is poor, leading to a low single-pass utilization (SPU) of just 5%, despite a high initial FE for CO. As the CO_2 flow rate is reduced to 10 ml/min, CO selectivity improves significantly, with FE(CO) remaining high, while the SPU increases to 62%. Further reducing the flow rate to 7.5 ml/min enhances the system efficiency even more, reaching an SPU of 90%, indicating optimal CO_2 utilization. The cell voltage remains relatively stable across all cases, though lower flow rates contribute to slightly lower voltages, suggesting improved energy efficiency. These results suggest that lowering the CO_2 flow rate enhances CO_2-to-CO conversion efficiency with a minimal FE(CO) drop from 98% to 88%. A key tradeoff between SPU and system stability emerges when optimizing CO_2 flow rates in electrolysis. While reducing the CO_2 flow rate significantly improves SPU—as seen in the 90% SPU at 7.5 ml/min—it may also introduce mass transport limitations, potentially affecting long-term operational stability. Lower flow rates enhance CO_2 utilization, reducing unreacted CO_2 losses and improving CO selectivity, but they may also lead to localized CO_2 depletion, pH fluctuations, and electrode degradation over extended operation, which could negatively impact stability. Conversely, higher flow rates (e.g., 150 ml/min, where SPU is only 5%) ensure a continuous CO_2 supply, maintaining a more stable reaction environment but at the cost of inefficient CO_2 usage and increased energy losses. A balanced tradeoff would involve selecting a moderate CO_2 flow rate (e.g., 10–30 ml/min), where SPU remains high (~60–80%) while ensuring adequate CO_2 availability for prolonged and stable operation without excessive CO_2 wastage.

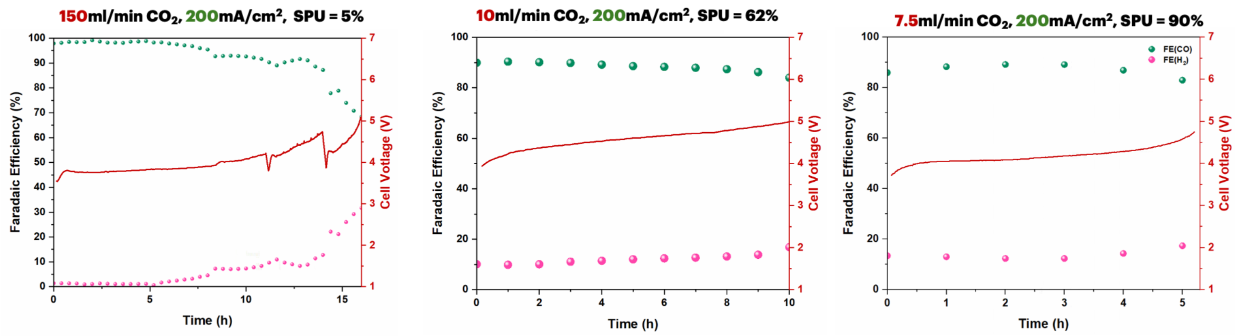

Figure 3. Long-term CO_2 electrolysis under different pure CO_2 flow rates (150, 10, and 7.5 ml/min) with 0.25M KOH and IrO_2-TiPt at 200mA/cm^2

4. Upcoming tasks

- Characterization of NiSACs
- Optimization of the 25cm^2 CO_2 electrolyzer operating condition:
    - Torque force
    - Cathodic/Anodic flow rate
    - Current density
    - Catalyst mass loading